



electricity my way

API Guide

Version: 0.9.8

Localvolts Pty Ltd

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Table 1: Version History

Version	Date	Change
0.9.8	15 th July 2026	Restructured guide with v2 as the primary interface and legacy v1 calls retained in Appendix A.
0.9.7	27 th May 2026	Updated from/to date section and /customer/metadata results.
0.9.6	27 th Feb 2026	Cut down document to cover only currently implemented functionality while other sections are being implemented and rewritten.
0.9.5	5 th March 2025	Added proposed functionality for v2 called including POST support and trade functions.
0.9.4	19 th Feb 2025	Updates to draft, moving v1 calls to Appendix.
0.9.3	12 th Feb 2025	Draft spec for v2 API added.
0.9.2	2 nd Sep 2024	Added note on Submitting Data - not yet implemented.
0.9.1	3 rd Nov 2023	Added Appendix for Home Assistant configuration.
0.9.0	16 th May 2023	Added optional from/to arguments and corrected intervalEnd description for /customer/interval. Removed dev-api. Removed /market/interval call. Added authentication error messages.
0.8.8	9 th May 2023	Switched interval end to UTC in /customer/interval. Version sync'd to API.
0.6.4	8 th May 2023	Added earnings and costs flex fields to /customer/interval.
0.6.3	1 st May 2023	Added 'N/A' to possible rate return values in /customer/interval.
0.6.2	4 th April 2023	Added /market/interval section.
0.6.1	23 rd March 2023	Updates to /customer/interval fields.
0.6.0	17 th March 2023	Initial Version.

1 Overview

The Localvolts API (Application Programming Interface) allows access to data from the Localvolts system without using the end-user interface.

The API is implemented as a stateless REST (REpresentational State Transfer) architecture and accessed via a HTTPS query string. The API supports 'GET' calls. The API is secured by TLS and authentication is performed using an API Key and 'Partner Id' in the HTTP header.

Most calls to the Localvolts API take the form:

```
https://<URL>/<API Version>/<Category>/<DataSet>[?<Arg1>=<value>, ...]
```

where the components of the call are explained in [table 2](#).



Table 2: Components of the API call

Component	Example	Description
<URL>	api.localvolts.com	URL of production API.
	api2.localvolts.com	URL of staging API.
	dev-api.localvolts.com	URL of dev/test API.
<API Version>		Version to support future versions with breaking changes and development preview versions.
	'v1', 'v2'	Major version of API, legacy call 'v1' still supported after the release of the 'v2' version of the call, which is not backwards compatible.
	'v101'	Development preview of version 1.0.1 of the call - one or more clients can upgrade to this call temporarily, and at a future stage this functionality will become the default 'v1' call.
<Category>	'customer'	Customer-specific data.
<DataSet>	'interval'	See section 1.3 .

1.1 Authentication

Authentication is required for every call to the API and comprises of an API key and Partner ID. The partner ID identifies who is making the call and the API Key is used to authenticate that user. There are two types of partner IDs, one for Localvolt's commercial partners who are managing groups of NMIs, and one for customer to access their own site information. Each type has an associated API Key. The former type are issued to the industry partner once an agreement with Localvolts has been reached. The latter can be accessed by a customer via the Localvolts web app (via the menu My Profile, General, API Key).

The API Key is very sensitive and therefore must be kept secret by the partner. A new key can be issued on request.

Authentication data is supplied by the caller in the HTTP header, and both the API key and partner Id must be provided. For example, using `curl` this would take the form:

```
curl -X GET https://api2.localvolts.com/v2/<Category/DataSet/Args>  
-H "Authorization: apikey <API Key>" -H "partner: <Partner ID>"
```

where:

<API Key> is a unique string generated by Localvolts
<Partner ID> is a unique industry partner or customer ID issued by Localvolts

If authentication fails, the API will return an error (HTTP status 500) with a message stating the reason:

No Authorisation header found	- No authorisation header was detected.
No API key provided	- No API key was found in the authorisation header.
No Partner Id provided	- No partner id was provided.
Unregistered partner: <partner id>	- the partner id was provided but not authorised.
Invalid API Key (partner: <partner id>)	- An invalid API key was provided for the partner.



1.2 Special Commands

The following commands do not take the form shown above include:

`<URL>/version` Returns the version number of the API as a JSON object
`{ version: <number> }`. Does not require authentication

1.3 Retrieving Data

A number of DataSets can be retrieved using the API. The DataSet is the type of data being requested within each Category (i.e. 'Market' or 'Customer'). Depending on which dataset is selected, there is a subset of valid arguments. The available datasets are shown in table [3](#).

Table 3: Data Sets

Category	DataSet	Description	Reference
customer	interval	Customer specific data for an interval range	3
customer	metadata	High level customer data such as LNSP and Tariff	4



2 Version 2 Overview

Version 2 of the Localvolts API introduces upgrades including:

- Splitting interval data into per-circuit (rather than per-site) rows (see [section 5](#));
- More flexible interval from/to parameters;
- New Demand charge cost and flex fields to accomodate complex and non-linear demand charge tariffs (see [section 5](#)).

2.1 HTTP Calls

HTTP call syntax and authentication is relatively unchanged, with the API Version component of the call URL simply being switched to 'v2'. A Partner Id and API Key are still required in the header data. Currently, the URL for the latest version is `api2.localvolts.com` rather than `api.localvolts.com`.



3 GET /customer/interval

Returns interval data for a specified customer or for all customers that a Partner is authorised to see. Note that authorisation of individual customers under a Partner ID is a process that occurs outside of the API.

By default, the call returns the current NEM (5 minute) period. Other periods can be retrieved using explicit interval timestamps.

3.1 Arguments

HTTPs GET query of the form /v2/customer/interval with authentication header and arguments as described in table 4.

Table 4: Arguments for GET /customer/interval

Argument	Example	Description
NMI=<NMI>	NMI=*	Return data for the specified NMI. Use '*' to request data for all NMIs that the Partner has been authorised to see. If absent, defaults to '*'.
from=<timestamp>	from=2023-05-15T02:40:00Z	Specifies the end of the first (5 minute) interval of the data set to be returned, as a ISO 8601 UTC date string. Note that if this date is not exactly the end of a NEM period, the period that contains this date will be returned as the first date. If absent, the starting date will be the current NEM period. Note that data can be requested only up to 72 hours in the past.
		Using the keyword 'current' means the current period (e.g. from=current will return only the current 5 minute interval).
to=<timestamp>	to=2023-05-15T02:45:00Z	Specify the end date for the data set, as a ISO 8601 UTC date string. If this date is not exactly the end of a NEM period, the period that contains this date will be returned as the final date. If absent, a single interval based on the 'to' argument will be returned. Data queried cannot exceed 24 hours into the future and there is a limit of 24 hours of data at a time for historical calls. This argument also supports keywords in the form of current, nDay(s) and nInterval(s), e.g. '1Day', '288Intervals', etc. This time will be added to the 'from' parameter.
		For Example: &from=current&to=current - return just the current period.
		&from=current&to=1day - return one day forward, starting at the current interval.
		&from=current&to=12intervals - return an hour forward starting at the current interval.



3.2 Examples

`https://api2.localvolts.com/v2/customer/interval?NMI=*`

or using curl (with authorisation header):

```
curl -X GET "https://api2.localvolts.com/v2/customer/interval?NMI=*"
-H "Authorization: apikey 62a72a83413a847be6c83156ac55bf00" -H "partner: 3268"
```

```
curl -X GET "https://api2.localvolts.com/v2/customer/interval?NMI=*&from=2023-05-15T02:40:00Z&to=2023-05-15T02:45:00Z" -H "Authorization: apikey 62a72a83413a847be6c83156ac55bf00" -H "partner: 3268"
```

3.3 Results

This returns a JSON structure comprising interval data in the form shown in table 5. Note that where multiple units are specified (e.g. ‘Wh’ or ‘kWh’, the initial unit will be the default unless optional scaling is added).

Table 5: Customer Interval Data

Field	Type	Value	Description
NMI	string	<NMI>	National Meter Identifier for which data is being provided. All data is that obtained at the NMI location. NMI will be a 10 character id, without checksum.
circuit	string	‘Import’, ‘Export’, etc.	The conceptual name of the circuit this data applies to.
			Import - main load
			Export - main export circuit
			CLoad - controlled load
			Import2 - secondary load
			etc.
register	string	examples: ‘E1’, ‘B1’, ‘11’, ‘61’	The meter register that this circuit is identified as.
direction	string	Buy or Sell	The direction of the amount or volume item. A positive value is in this direction (for example a positive amount is a cost if direction = Buy and revenue if direction = Sell.
intervalEnd	string	YYYY-MM-DDTHH:mm:ssZ	Time at the end of the NEM interval, in UTC (24-hour format). Note that the NEM always runs off AEST (UTC+10)
intervalDuration	number	5	Default for NEM is 5 minutes
intervalDurationUnits	string	‘minutes’	
volume	number	<variable>	all energy drawn from the grid or injected into the grid for the interval
volumeUnits	string	‘kWh’	or ‘Wh’ depending on scaling

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Field	Type	Value	Description
maxDemand	number	<variable>	max peak demand based on demand on current metered maximum that affects this interval.
demandUnits	string	'kW'	or 'kVA' depending on how peaks are used by LNSPs
amountAll	number	<variable>	all costs or earnings for the interval
amountAllUnits	string	'cents'	or '\$' depending on scaling
spotCost	number	<variable>	wholesale settlement cost
spotCostUnits	string	'cents'	or '\$' depending on scaling
matchedCost	number	<variable>	peer-to-peer settlement cost
matchedCostUnits	string	'cents'	or '\$' depending on scaling
amountVar	number	<variable>	all variable costs or earnings for the interval
amountVarUnits	string	'cents'	or '\$' depending on scaling
amountFixed	number	<variable>	all fixed costs or earnings for the interval
amountFixedUnits	string	'cents'	or '\$' depending on scaling
amountDemand	number	<variable>	demand charge, attributed to each interval where demand window is in effect (e.g. monthly charge split across all peak demand intervals in the month).
amountDemandUnits	string	'cents'	or '\$' depending on scaling
rateAllVar	number or string	<variable>	variable cost or earnings for the interval, expressed as a rate. Note that when there are no volume, this value is 'N/A'
rateAllVarUnits	string	'c/kWh'	or '\$/kWh' or '\$/MWh' depending on scaling
proportionP2P	number	<variable>	The proportion of volume that is covered by peer-to-peer trades. This is a number between 0 and 1 (i.e. 0%-100% of volume). Note this figure is the current expected settlement (taking into account counterparty positions) and not the original dealt figure.
flexUp	number	<variable>	The expected change in cost or earnings when importing or exporting an additional amount of energy from/to the grid. Multiplying this field by the proposed export volume (in kWh) will give the expected change in costs or earnings.
flexDown	number	<variable>	The expected change in cost or earnings when reducing volume on this circuit. Multiplying this field by the proposed decrease in volume (in kWh) will give the expected change in cost or earnings.
flexUnits	string	'c/kWh'	
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Field	Type	Value	Description
flexDemandUp	number	<variable>	impact on costs for this interval if consumption is increased by 1 demand unit (see <i>demandUnits</i>). Note that increasing max demand will also impact other intervals in the relevant period (potentially monthly or annually depending on demand or capacity charge), therefore the monthly sum of flexDemandUp will give the monthly impact of flexing a single interval in a demand period by 1 <i>demandUnits</i> above <i>maxDemand</i> . If there is no demand charge or we are not in a demand metering window, flexDemandUp will equal 0.
flexDemandDown	number	<variable>	impact on costs for this interval if consumption is reduced by 1 demand unit (see <i>demandUnits</i>).
flexDemandUnits	string	'c/kW/Day'	Note, may expressed in '\$/kW/Month' for some LNSPs (or '\$/kVA/Month')
emissions	number	<variable>	emissions included in all energy drawn from the grid for the interval
emissionsUnits	string	'g-CO2e'	Note, may switch to kg-CO2e depending on scaling of main field
zeroEE	number	<variable>	proportion of zero emissions energy draw from / injected into the grid for the interval
zeroEEUnits	string	'%'	
quality	string		Quality indicator for all data in this record.
		'Act'	All source data used in the calculations is actual. This would only be the case for historical queries.
		'Sub'	Meter data used was substituted by the meter data provider.
		'FSub'	Meter data used was a final substitution by the meter data provider.
		'Exp'	The expected value - usually when we have some actual elements (such as current period price) and other forecast elements (such as LGC price, level of demand or demand charge);
		'Fcst'	All of the time-dependent elements of the calculation are forecasts.
lastUpdate	string	YYYY-MM-DDTHH:mm:ssZ	Date/time this data-set was updated, in UTC.



4 GET /customer/metadata

Returns metadata data for a specified customer or for all customers that a Partner is authorised to see.

4.1 Arguments

HTTPs GET query of the form /v2/customer/metadata with authentication header and arguments as described in table 6.

Table 6: Arguments for GET /customer/metadata

Argument	Example	Description
NMI=<NMI>	NMI=*	Return data for the specified NMI. Use '*' to request data for all NMIs that the Partner has been authorised to see. If absent, defaults to '*'. NMI = National Meter Identifier. NMI will be a 10 character id, without checksum.

4.2 Examples

`https://api2.localvolts.com/v2/customer/metadata?NMI=4100000000`

or using curl (with authorisation header):

```
curl -X GET "https://api2.localvolts.com/v2/customer/metadata?NMI="
-H "Authorization: apikey <API Key>" -H "partner: <Partner ID>"
```

4.3 Results

This returns a JSON structure comprising interval data in the form shown in table 7. Note that where multiple units are specified (e.g. 'Wh' or 'kWh', the initial unit will be the default unless optional scaling is added).

Table 7: Customer Metadata

Field	Type	Example	Description
NMI	string	<NMI>	National Meter Identifier for which data is being provided.
Region	string	'NSW', 'QLD', etc.	NEM region (wholesale price reference).
Circuit	string	'Import', 'Export', etc.	The conceptual name of the circuit this data applies to.
Suffix	string	'E1', 'B1', etc.	The datastream suffix from the meter.
Jurisdiction	string	'ACT', 'NSW', etc.	Jurisdiction governing state-based schemes and rules.

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Field	Type	Example	Description
LNSP	string	Ausgrid	Local Network Service Provider. The organisation that owns the poles and wires used to distribute electricity to premises within a geographical area and in the relevant participating jurisdiction.
DLF	string	'JL40'	Distribution Loss Factor code.
TNI	string	'NSN1'	Transmission Node Identifier - a four-character code conceptually representing a Transmission Connection Point, where the Distribution Network meets the Transmission Network.
Tariff	string	'EA116', 'EA029', etc.	Tariff code for this circuit.
MDP	string	'Bluecurrent (Australia) Pty Ltd'	Meter data provider.
ReadType	string	'Remote Interval', 'Manual Interval' or 'Accumulation'	Meter read type.



5 Contact

For technical support or other requests, please contact Localvolts Pty Ltd:

<https://localvolts.com>

support@localvolts.com

Phone: +61-2-8006 8052



6 Appendix A - Legacy Calls From v1

This section lists the original calls from version 'v1' of the API. These calls remain supported in production, however new integrations should use the v2 calls described in the main body of this guide. Note that the 'POST' calls originally proposed for v1 were never implemented and have been removed from this guide.

6.1 GET /customer/interval

Returns interval data for a specified customer or for all customers that a Partner is authorised to see. Note that authorisation of individual customers is a process that occurs outside of the API.

By default, the call returns the current NEM (5 minute) period, plus all periods to 24hrs from the current period. Support for historical and other forward periods is available upon application to Localvolts..

Arguments and Examples

Table 8 shows the supported arguments:

Table 8: Arguments for GET /customer/interval

Argument	Example	Description
NMI=<NMI>	NMI=*	Return data for the specified NMI. Use '*' to request data for all NMIs that the Partner has been authorised to see. If absent, defaults to '*'.
from=<timestamp>	from=2023-05-15T02:40:00Z	Specifies the starting date of the data set to be returned, as a ISO 8601 UTC date string. Note that if this date is not exactly the end of a NEM period, the period that contains this date will be returned as the first date. If absent, the starting date will be the current NEM period. Note that data can be requested only up to 72 hours in the past.
to=<timestamp>	to=2023-05-15T02:45:00Z	Specify the end date for the data set, as a ISO 8601 UTC date string. If this date is not exactly the end of a NEM period, the period that contains this date will be returned as the final date. If absent, the 24 hours of data following the starting date will be returned. Data queried cannot exceed 24 hours into the future and there is a limit of 24 hours of data at a time for historical calls.

Example:

```
https://api.localvolts.com/v1/customer/interval?NMI=*
```

or using curl (with authorisation header):

```
curl -X GET "https://api.localvolts.com/v1/customer/interval?NMI=*"
-H "Authorization: apikey 62a72a83413a847be6c83156ac55bf00" -H "partner: 3268"
```



```
curl -X GET "https://api.localvolts.com/v1/customer/interval?NMI=*&from=2023-05-15T02:40:00Z&to=2023-05-15T02:45:00Z" -H "Authorization: apikey 62a72a83413a847be6c83156ac55bf00" -H "partner: 3268"
```

Results

This returns a JSON structure comprising interval data in the form shown in table 9. Note that where multiple units are specified (e.g. ‘Wh’ or ‘kWh’, the initial unit will be the default unless optional scaling is added).

Table 9: Customer Interval Data

Field	Type	Value	Description
NMI	string	<NMI>	National Meter Identifier for which data is being provided. All data is that obtained at the NMI location. NMI will be a 10 character id, without checksum.
intervalDuration	number	5	Default for NEM is 5 minutes
intervalDurationUnits	string	‘minutes’	
intervalEnd	string	YYYY-MM-DDTHH:mm:ssZ	Time at the end of the NEM interval, in UTC (24-hour format). Note that the NEM always runs off AEST (UTC+10)
exportsAll	number	<variable>	all energy injected into the grid for the interval
exportsAllUnits	string	‘Wh’	or ‘kWh’ depending on scaling
importsAll	number	<variable>	all energy drawn from the grid for the interval, including controlled loads
importsAllUnits	string	‘Wh’	or ‘kWh’ depending on scaling
demandMain	number	<variable>	peak average demand used to calculate demand charges for primary load circuit only (demand charges are not applied to controlled load circuits)
demandMainUnits	string	‘kW’	or ‘kVA’ depending on how peaks are used by LNSPs
demandPeriod	number	30	period of time over which peak demand is averaged to obtain demandMain
demandPeriodUnits	string	‘minutes’	
demandInterval	boolean	0 or 1	0: interval is not relevant for peak average demand based on which demand charges are determined 1: interval is relevant for peak average demand based on which demand charges are determined
earningsAll	number	<variable>	all earnings for the interval
earningsAllUnits	string	‘cents’	or ‘\$’ depending on scaling
earningsAllVar	number	<variable>	all variable earnings for the interval
earningsAllVarUnits	string	‘cents’	or ‘\$’ depending on scaling
earningsAllFixed	number	<variable>	all fixed earnings for the interval
earningsAllFixedUnits	string	‘cents’	or ‘\$’ depending on scaling
earningsAllVarRate	number or string	<variable>	variable earnings for the interval, expressed as a rate. Note that when there are no exports, this value is ‘N/A’

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6 Appendix A - Legacy Calls From v1

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Field	Type	Value	Description
earningsAllVarRateUnits	string	'c/kWh'	or '\$/kWh' or '\$/MWh' depending on scaling
earningsFlexUp	number	<variable>	The expected change in earnings when exporting an additional amount of energy to the grid. Multiplying this field by the proposed export volume (in kWh) will give the expected change in earnings.
earningsFlexDown	number	<variable>	The expected change in earnings when reducing exports to the grid. Multiplying this field by the proposed decrease in exports (in kWh) will give the expected change in earnings.
earningsFlexUnits	string	'c/kWh'	
costsAll	number	<variable>	all costs for the interval for all circuits
costsAllUnits	string	'cents'	or '\$' depending on scaling
costsAllVar	number	<variable>	all variable costs for the interval for all circuits, excluding demand costs
costsAllVarUnits	string	'cents'	Note, may expressed in '\$' depending on scaling of main field
costsAllFixed	number	<variable>	all fixed costs for the interval for all circuits
costsAllFixedUnits	string	'cents'	or '\$' depending on scaling
costsDemandMain	number	<variable>	all demand costs for the interval for main circuit
costsDemandMainUnits	string	'c/kW/Day'	Note, may expressed in '\$/kW/Month' for some LNSPs (or '\$/kVA/Month')
costsAllVarRate	number or string	<variable>	all variable costs' rate for the interval for all circuits, excluding demand costs. Note that when there are no imports, this rate is 'N/A'
costsAllVarRateUnits	string	'c/kWh'	Note, may switch to '\$/kWh' or '\$/MWh' depending on scaling of main field
costsFlexUp	number	<variable>	The expected change in costs when importing an additional amount of energy from the grid. Multiplying this field by the proposed import volume (in kWh) will give the expected overall change in cost.
costsFlexDown	number	<variable>	The expected change in costs when reducing imports from the grid. Multiplying this field by the proposed import reduction (in kWh) will give the expected change in overall cost.
costsFlexUnits	string	'c/kWh'	
exportsAllEmissions	number	<variable>	emissions included in all energy injected into the grid for the interval
exportsAllEmissionsUnits	string	'g-CO2e'	Note, may switch to kg-CO2e depending on scaling of main field
importsAllEmissions	number	<variable>	emissions included in all energy drawn from the grid for the interval, including controlled load
importsAllEmissionsUnits	string	'g-CO2e'	Note, may switch to kg-CO2e depending on scaling of main field

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Field	Type	Value	Description
exportsAllZeroEE	number	<variable>	proportion of zero emissions energy injected into the grid for the interval
exportsAllZeroEEUnits	string	'%'	
importsAllZeroEE	number	<variable>	proportion of zero emissions energy drawn from the grid for the interval
importsAllZeroEEUnits	string	'%'	
quality	string	'Act', 'Exp' or 'Fcst'	Quality indicator for all data in this record. Act: All source data used in the calculations is actual. This would only be the case for historical queries. Exp: The expected value - usually when we have some actual elements (such as current period price) and other forecast elements (such as LGC price, level of demand or demand charge); Fcst: All of the time-dependent elements of the calculation are forecasts.
lastUpdate	string	YYYY-MM-DDTHH:mm:ssZ	Date/time this data-set was updated, in UTC.



7 Appendix B - Home Assistant

The example YAML configuration below can be used to point Home Assistant at the Localvolts API. Please note this solution has not been tested by Localvolts nor is it warranted to work or produce correct results.

```
# Example configuration.yaml entry
rest:
  - authentication: basic
    scan_interval: 20
    resource: "https://api.localvolts.com/v1/customer/interval?NMI=*"
    headers:
      Authorization: apikey <YourAPIKeyHere>
      partner: <YourPartnerId>
      User-Agent: Home Assistant
    sensor:
      - name: "Localvolts"
        json_attributes_path: "$.[0]"
        value_template: >
          {% if (value_json.0.quality == 'Exp') %}
            {{value_json.0.earningsFlexUp | round(4) }}
          {% else %}
            {{ states('sensor.Localvolts') }}
          {% endif %}
        json_attributes:
          - "intervalEnd"
          - "earningsFlexUp"
          - "costsFlexUp"
          - "quality"
        unit_of_measurement: "$/kWh"
```

Note the following:

- Replace <YourAPIKeyHere> with the API key that was sent to you. Leave 'apikey' before it;
- Replace <YourPartnerId> with the Partner Id that was sent to you;
- Refer to table 9 of this guide for a description of fields such as *earningsFlexUp* and *costsFlexUp*;
- The 'scan_interval' is set to 20 seconds in this example. Although data from the API will be updated every 5 minutes, the settlement price for an interval is published by AEMO a short time (around 20 seconds) after the beginning of an interval. Sampling every 20 seconds allows this price to be picked up;
- The test of the *quality* field is to determine whether the final settlement price has been received yet for this interval. Refer to the field definition in table 9 of this guide;